Literature Review

Machine Learning for Food Waste Risk Prediction in Perishable Retail

1. Introduction and Scope

This review surveys research and industry practice relevant to a project that classifies the waste risk of perishable products (LOW, MEDIUM, HIGH) and recommends an action (normal selling, a 10-15% discount, or a 25% discount plus donation to an NGO). The project dataset contains 1,000 records for eight products (eggs, rice, bread, chicken, tomato, banana, apple, milk) with features such as quantity, price, expiry days, temperature, day of week and predicted demand. The review is organized around four themes: the scale of the problem, demand forecasting for perishables, markdown and pricing decisions, and the machine-learning methods and evaluation practices needed to build a reliable classifier.

Note on sources: web-based sources were read at abstract or summary level; full texts were not reviewed. Please verify each reference and add page numbers or DOIs before submission.

2. The Food Waste Problem

The UNEP Food Waste Index Report 2024 estimates that 1.05 billion tonnes of food were wasted in 2022, about 19% of the food available to consumers. Households accounted for roughly 60% of this total, food services for 28%, and retail for 12% (about 131 million tonnes) [1]. Retail is therefore a smaller share than households, but it is the point where purchasing, pricing and stock rotation decisions are made, and where data are most readily available. The report also notes a direct relationship between average temperatures and food waste levels [1], which supports including temperature as a predictive feature in this project. It further observes that many low- and middle-income countries lack adequate systems for tracking retail and food-service waste [1], which helps explain why open datasets in this area are scarce.

3. Demand Forecasting for Perishable Products

A large part of the applied literature treats waste as a downstream effect of inaccurate demand forecasts and over-ordering. A prototype developed in the APPETITE research project uses machine-learning models to predict daily sales and feeds those forecasts into order optimization, on the premise that retailers' over-ordering is a direct cause of expired or spoiled stock [2]. A literature review on data-driven approaches in supermarkets likewise frames demand prediction as the main prevention stage for perishable waste, and points out that categories such as fruit and vegetables, dairy and poultry have short shelf lives and therefore contribute disproportionately to waste [4].

Albu [3] examined daily sales from the Corporacion Favorita dataset (three stores, ten perishable product families, 2016) and focused on over-forecast errors, where predicted demand exceeds actual sales. Promotions and holidays both increased these errors, with holidays causing the greatest instability. Random Forest and XGBoost reduced the magnitude of over-forecasting relative to a seasonal naive benchmark. Importantly, the naive model scored best on MAPE while XGBoost produced the lowest estimated waste, so conventional accuracy metrics and waste-oriented metrics can lead to different conclusions [3]. This finding is directly relevant to the present project: the evaluation metric must reflect the cost of missing a high-risk item, not only overall accuracy.

4. Risk Classification and Early-Warning Approaches

Forecasting predicts how much will sell; a risk classifier predicts whether a specific stock position is likely to become waste and what should be done about it. The searches conducted for this review found far fewer peer-reviewed studies that frame the problem as a multi-class risk label than studies on demand

forecasting or pricing. Industry practice is moving in this direction, with commercial tools offering predictive markdown prompts ahead of expiry, but these are vendor materials rather than evaluated research. This gap between forecasting and decision-oriented classification is where the present project is positioned.

5. Markdown Pricing and Redistribution

Once risk is identified, the remaining question is which action to take. Discounting near-expiry products is the most common lever. Wasteless, a start-up supported by EIT Food, applies expiry-based dynamic markdowns rather than a single last-day discount, using reinforcement learning to evaluate and refine its pricing decisions; the reasoning is that shoppers tend to prefer items with later expiry dates, so late discounts are often too small to move stock [5]. At Albert Heijn, markdown level selection was modelled as a Markov Decision Process, and offline Q-learning on historical data was used to learn a policy that minimizes waste without greatly increasing markdown cost [6]. An academic study proposed a dynamic markdown pricing model for perishables using Deep Q-Networks, which considers freshness, customer characteristics and inventory level, and compared it against two-period pricing and random markdown baselines [7].

These approaches optimize price continuously and require rich transaction or customer data. The project dataset, by contrast, uses three discrete action tiers, which resembles a rule-based markdown policy. A supervised classifier that recommends one of these tiers is a simpler and more transparent alternative, although it cannot learn the price-response behavior that the reinforcement-learning systems target.

6. Machine-Learning Methods and Evaluation

Models. Tree-based ensembles are a natural fit for small tabular datasets with mixed feature types. Random Forests average many decorrelated trees and are robust to noise [8]. Gradient-boosted trees such as XGBoost [9] and LightGBM [10] typically achieve higher accuracy through sequential error correction, and XGBoost has already shown benefits in the perishable-waste setting [3].

Class imbalance. The dataset is imbalanced (MEDIUM is only 16% of records). Surveys of learning from imbalanced data recommend resampling, cost-sensitive learning and appropriate metrics instead of accuracy [12]. SMOTE generates synthetic minority examples by interpolating between neighbors [11], while class weights are a simpler alternative that avoids creating artificial records.

Interpretability. For a decision-support tool, staff need to understand why an item is flagged. SHAP assigns each feature an additive contribution to an individual prediction and provides a consistent framework for explaining tree ensembles [13].

Data leakage and validation. Kaufman et al. [14] describe leakage as the introduction of information about the target that would not be available at prediction time, a common cause of unrealistically high performance. In this project, if risk_level was generated by rules over expiry_days, quantity and predicted_demand, a model could simply recover the rule. Chronological train/test splits, a rule-based baseline for comparison, and per-class metrics (especially recall for HIGH) are the appropriate safeguards.

7. Summary of Key Studies

Source	Focus	Method	Main finding	Relevance
[1] UNEP 2024	Global waste estimate	Data synthesis	1.05 Bt wasted in 2022; retail 12%; temperature linked to waste	Motivation; temperature feature
[2] APPETITE	Ordering and forecasting	ML sales forecasts	Over-ordering drives retail waste; forecasts feed order optimization	Problem framing

Source	Focus	Method	Main finding	Relevance
[3] Albu	Over-forecast risk	RF, XGBoost vs seasonal naive	XGBoost lowest estimated waste despite worse MAPE	Metric choice
[4] Review	ML for perishable waste	Literature review	Demand prediction central to prevention	Background
[5] Wasteless	Expiry-based markdown	Dynamic pricing + RL	Earlier, graduated discounts outperform last-day stickers	Action design
[6] Albert Heijn	Markdown policy	MDP + offline Q-learning	Learns markdown policy from history	Decision tiers
[7] Ewha study	Perishable pricing	Deep Q-Network	Compared with two-period and random markdown	Alternative approach

8. Research Gap and Positioning

The literature shows strong work on demand forecasting and on price optimization, but fewer openly documented studies that classify waste risk into actionable tiers. Existing work also stresses that accuracy metrics can mislead, and that models must be judged by their waste-related consequences [3]. The present project addresses this by (a) formulating a three-class risk problem tied to concrete actions, (b) using a time-based split and a rule-based baseline to test for leakage, (c) emphasizing recall on the HIGH class, and (d) explaining predictions with SHAP. A key limitation is the small, likely synthetic dataset covering one six-month period; results should be presented as a proof of concept rather than evidence of real-world performance.

References

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